DriveBench: Offset-Free MPC and Integral-Augmented LQR for an Industrial DC Motor Drive, Compared with Hand-Tuned and Optimised PID

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Abstract

Model predictive control (MPC) plans the next actuator move by rolling a plant model forward over a finite horizon and solving a constrained optimisation at every sample. We build an offset-free MPC and an integral-augmented linear quadratic regulator (LQR) for the speed of a permanent-magnet DC (PMDC) motor, and compare them with a hand-tuned PID and an optimised PID on ten reproducible scenarios. All four controllers command the armature voltage through the same simulated drive, which saturates the voltage, limits its slew rate, and protects against over-current, and they face identical references, load torques and noise. The MPC predicts armature current and speed, respects voltage, voltage-rate and current constraints, and estimates both the load torque and an armature-voltage disturbance so that it stays offset-free under resistance mismatch. Optimised PID gains are found once on a speed-tracking scenario and then frozen. MPC has the lowest speed-tracking error and the fastest settling in most scenarios, including tracking, encoder noise, reversal and plant mismatch, and the lowest time spent in current limit under current protection; the LQR is best on startup settling and on load disturbances. Where the drive limits dominate, the four controllers are close. The MPC and LQR also use the measured current, which the PID loops do not, their weights were set by hand while looking at these scenarios, and all parameters are illustrative.

1 Introduction

A PMDC motor with controlled armature voltage is a standard industrial drive: it turns conveyors, pumps, machine axes and material-handling equipment. The control task is to hold a shaft speed while the load changes, the supply wobbles and the encoder is noisy, without asking the drive for more voltage, voltage slew or current than it may deliver. A PID loop reacts only after the speed has already moved, and a limit applied after the fact does not undo the mistake that the controller made while it did not know about the limit. MPC uses a model to predict the response to different voltage choices, re-solves every sample (receding horizon), and can enforce voltage, rate and current limits inside the optimiser [2, 4].

We compare it with a linear quadratic regulator with integral action (LQI) [3] and two PID loops. We implement MPC with a two-state linear prediction model, a disturbance estimate for offset-free tracking [1], a constrained quadratic programme solved with SLSQP, and a simulated motor as the truth plant. All motor numbers are illustrative, not data from a commercial motor.

2 Background and models

We keep four models separate: the actual plant, the controller's prediction model, the drive, and the measurement model.

2.1 Actual plant

The plant is the coupled electrical and mechanical system

$$L_a \frac{di_a}{dt} = V_a - R_a i_a - K_e \omega, \qquad J \frac{d\omega}{dt} = K_t i_a - B\omega - T_L, \qquad \frac{d\theta}{dt} = \omega, \tag{1}$$

with R_a=1.0 Ω, L_a=10 mH, K_t=K_e=0.05, J=0.002 kg m^2, B=0.001 N m s, nominal load T_L=0.05 N m, $V_{\max}$=24 V, $I_{\max}$=8 A. In state-space form with $x = [i_a, \omega, \theta]^\top$, $A = \left[\begin{smallmatrix} -100 & -5 & 0 \\ 25 & -0.5 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right]$, $B = [100, 0, 0]^\top$, $E = [0, -500, 0]^\top$, and the poles lie at -98.7 and $-1.75\,\mathrm{s}^{-1}$: an electrical time constant of 10 ms and a mechanical one of about 0.57 s. The plant is integrated with fixed-step RK4 at 1 ms while the controller runs every 10 ms with its output held; a 10 ms integration step would not resolve the electrical mode.

2.2 Prediction model and observer

MPC predicts with the 2-state model $x_{k+1} = A_d x_k + B_d(V_k + \hat{d}_V) + E_d \hat{d}_T$, $x = [i_a, \omega]^\top$, obtained by exact zero-order-hold discretisation of the nominal motor. Each sample the innovations of the measured current and speed are mapped back to two disturbance estimates through $[B_d\ E_d]$, scaled by an observer gain g=0.1. The torque estimate $\hat{d}_T$ absorbs load and friction error; the voltage estimate $\hat{d}_V$ absorbs electrical error such as a wrong resistance. A torque-only estimate left a speed offset of about 44 RPM under +20% resistance in development, which estimating both removed [1].

2.3 Drive

Every command passes through one drive: enable and latched emergency stop, a slew limit ($1200\,\mathrm{V\,s}^{-1}$), saturation at $\pm V_{\max}$ times the DC-bus level, and current protection. Protection is a voltage window that keeps the exact one-sample electrical response inside $\pm I_{\max}$, computed from the drive's nominal R_a, K_e and the measured current and speed, with a feedback term on measured excess current; it is not a clip of the simulated state.

2.4 Measurement

Speed comes from an encoder with seeded Gaussian noise; current is measured ideally.

3 Controllers

3.1 MPC

Over N=60 samples (0.6 s) the controller chooses n_u=6 voltage moves with block lengths $1, 2, 4, \ldots$ samples and minimises

$$J = q \sum_{k=1}^{N} (\omega_k - r_k)^2 + q_T (\omega_N - r_N)^2 + r_\Delta \sum_j \Delta V_j^2, \tag{2}$$

in RPM^2 and V^2, with reference preview r_k, q=1, q_T=50, r_Δ=5000. Constraints are the voltage box, the voltage change between moves, and the predicted armature current $|i_{a,k}| \le I_{\max}$ over the whole horizon. All are linear in the decision variables, so SLSQP receives analytic gradients; solves take about 1–5 ms.

3.2 LQR with integral action

With $\dot{\xi} = \omega - r$ the augmented state is $[i_a, \omega, \xi]^\top$. The plant part is discretised exactly (zero-order hold) at the 10 ms controller period, the integrator uses the same forward-Euler update the controller

runs, and the gain $K = (R + B_d^\top P B_d)^{-1} B_d^\top P A_d$ comes from the discrete algebraic Riccati equation. The cost uses normalised weights q_i=0.1, q_ω=300, q_ξ=10^4, r_V=1 on states scaled by $I_{\max}$, $\omega_{\max}$, 0.5 s $\omega_{\max}$ and $V_{\max}$. The law is $u = u_{ff}(r) - K_x(x - x_{eq}(r)) - K_\xi \xi$, with back-calculation anti-windup against the applied voltage; a constant load is rejected by the integrator, with no disturbance estimator. The weights were chosen from a small sweep on startup and are hand-set, like the MPC settings.

3.3 PID baselines

The positional PID takes RPM and returns volts, with a filtered derivative on the measurement and back-calculation anti-windup against the voltage the drive *actually applied*, so slew, saturation and current limiting all unwind the integrator. Hand-tuned gains (K_p=0.01, K_i=0.03, K_d=0) are a deliberately conservative guess. Optimised gains minimise a dimensionless sum of IAE, overshoot, settling time, voltage effort, time in voltage saturation and time in current limit with L-BFGS-B [5]. They are found once on the speed-tracking scenario (K_p=0.0326, K_i=0.131, K_d=0) and frozen for all comparisons.

4 System design and implementation

Figure 1 shows the closed loop. All controllers drive the same plant through the same drive and are scored with the same metrics.

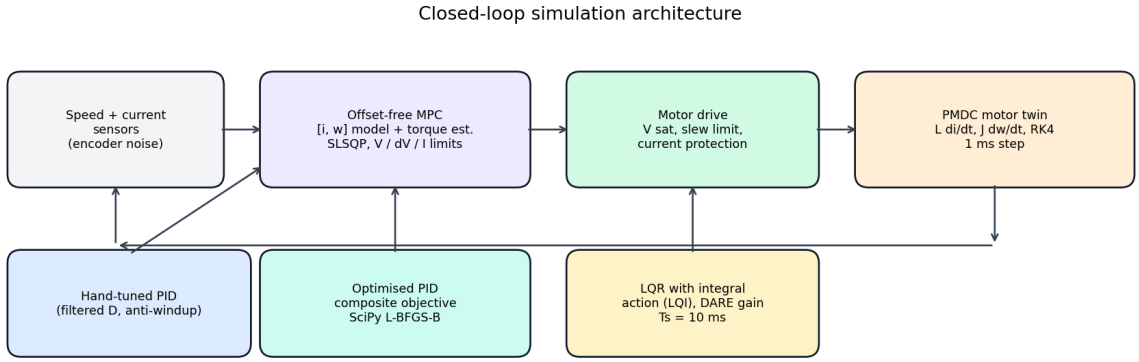


Figure 1: Simulation architecture: PID, optimised PID, LQR and offset-free MPC act on one motor through one drive.

5 Experimental setup

Controller sample 10 ms, plant step 1 ms, nominal load 0.05 N m. The ten scenarios, each reproducible (seeded noise, identical arrays for every controller), are:

1. **Motor startup:** 0 → 1000 RPM at nominal load.
2. **Speed tracking:** 1000/2000/1500/2500 RPM, 5 s each.
3. **Sudden load:** at 2000 RPM, load 0.05 → 0.10 N m at 5 s.
4. **Load removal:** 0.10 → 0.05 N m at 5 s.
5. **Voltage saturation:** 2900 RPM reference, DC bus sags to 85% at 8 s.
6. **Current protection:** 2500 RPM, load rises to 0.14 N m, which needs more than 8 A to hold speed.

7. **Model mismatch:** plant R_a+20%, J+30%, B+25% (and the opposite corner), load step at 8 s; controllers keep the nominal model.
8. **Sensor noise:** 15 RPM encoder noise during a 1000 → 1500 RPM step.
9. **Variable load:** 0.5 Hz sinusoid plus bounded random torque at 1500 RPM.
10. **Reverse operation:** +1000 → −1000 → +1000 RPM.

Settling uses a 2% band on the last reference segment; IAE is in RPM s.

6 Results

Blue bold cells mark where MPC is lowest (or tied lowest) among the four controllers on settling time, overshoot or IAE; black bold marks another controller that is lowest or tied. Peak current and time in current limit are shown for context, not ranked. All numbers are generated by `scripts/rebuild_report_metrics.py` from the code in this repository.

Table 1: Closed-loop metrics for the ten scenarios (plus the opposite mismatch corner). Optimised PID uses the frozen gains.

Scenario	Controller	Settling [s]	Overshoot [%]	IAE [RPM s]	Peak I [A]	Current-limited [s]
Motor startup	PID	1.84	2.8	414	7.97	0.20
	Opt. PID	0.92	0.2	364	7.99	0.45
	LQR	**0.71**	**0.0**	**346**	7.99	0.67
	MPC	0.76	**0.0**	351	7.98	0.60
Speed-reference tracking	PID	1.58	1.0	1854	7.99	1.81
	Opt. PID	1.49	0.1	1660	7.99	2.65
	LQR	1.46	**0.0**	1608	7.99	3.42
	MPC	**0.87**	**0.0**	**1071**	7.99	3.05
Sudden mechanical load	PID	**0.00**	**0.0**	35	6.33	0.00
	Opt. PID	**0.00**	**0.0**	8	6.35	0.00
	LQR	**0.00**	**0.0**	**1**	6.35	0.00
	MPC	**0.00**	**0.0**	3	6.42	0.00
Load removal	PID	**0.00**	**0.0**	35	6.19	0.00
	Opt. PID	**0.00**	**0.0**	8	6.19	0.00
	LQR	**0.00**	**0.0**	**1**	6.19	0.00
	MPC	**0.00**	**0.0**	3	6.19	0.00
Voltage saturation	PID	—	0.3	4122	7.99	3.00
	Opt. PID	—	**0.0**	4092	8.00	3.53
	LQR	—	**0.0**	4072	8.00	3.98
	MPC	—	**0.0**	**3307**	8.00	3.84
Current protection	PID	3.92	0.2	3678	8.00	9.18
	Opt. PID	**3.89**	**0.0**	3635	8.00	9.71
	LQR	**3.89**	**0.0**	**3616**	8.00	10.05
	MPC	**3.89**	**0.0**	3626	8.00	4.34
Model mismatch	PID	2.69	1.4	1794	7.23	4.10
	Opt. PID	2.67	0.4	1700	7.36	4.25
	LQR	2.67	**0.2**	1676	7.36	4.36
	MPC	**2.08**	0.3	**1290**	7.14	4.47
Sensor noise	PID	0.68	5.0	181	7.98	0.08
	Opt. PID	0.63	**1.8**	160	7.99	0.22
	LQR	0.46	3.9	173	7.99	0.54
	MPC	**0.25**	2.3	**103**	7.98	0.16
Variable load	PID	11.74	**0.0**	275	5.32	0.00
	Opt. PID	**0.00**	**0.0**	96	5.50	0.00
	LQR	**0.00**	**0.0**	**18**	5.44	0.00
	MPC	**0.00**	**0.0**	45	5.48	0.00
Reverse operation	PID	2.37	1.4	2199	7.97	1.10
	Opt. PID	1.44	0.1	2079	7.97	1.65
	LQR	1.24	**0.0**	2037	7.98	2.10
	MPC	**0.69**	**0.0**	**1132**	7.98	1.62
Model mismatch (low R, J, B)	PID	1.17	**0.0**	756	9.10	0.25
	Opt. PID	0.87	**0.0**	615	9.10	0.50
	LQR	0.57	**0.0**	518	9.10	0.95
	MPC	**0.33**	**0.0**	**468**	8.98	0.39

Figure 2 summarises IAE and Figure 3 shows four responses.

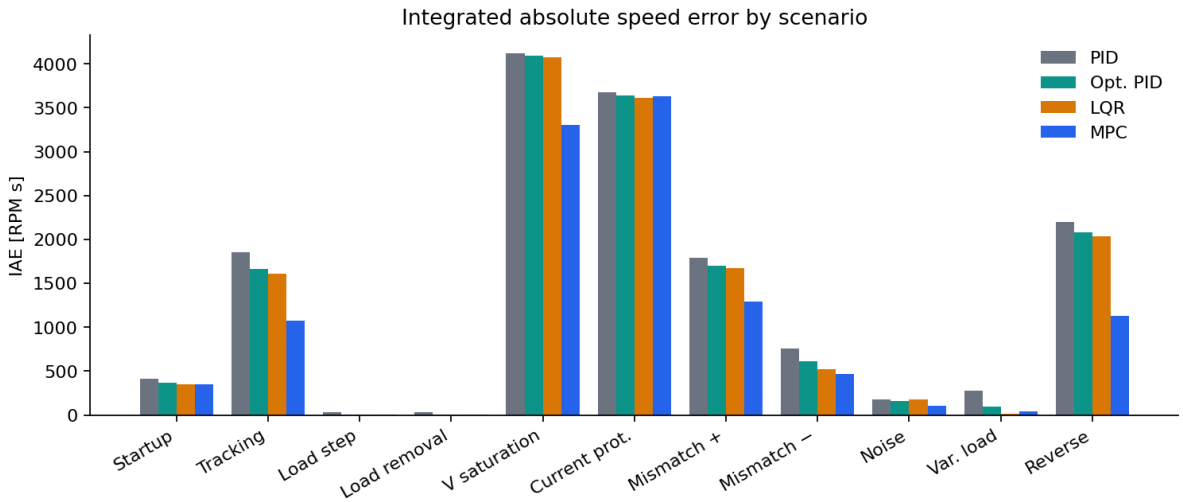
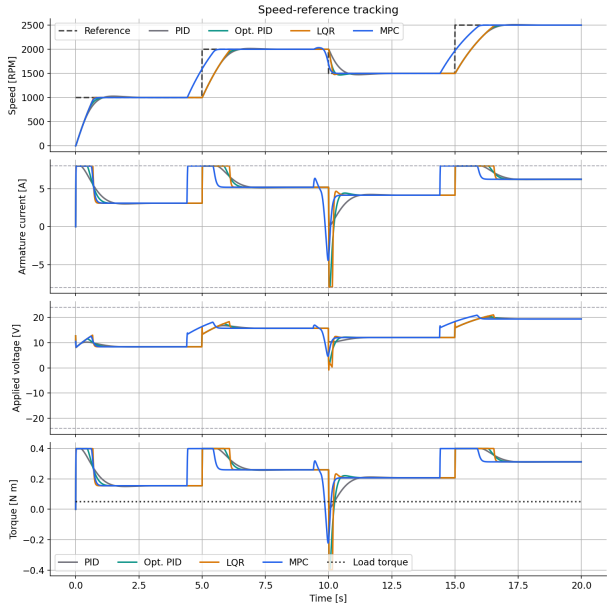
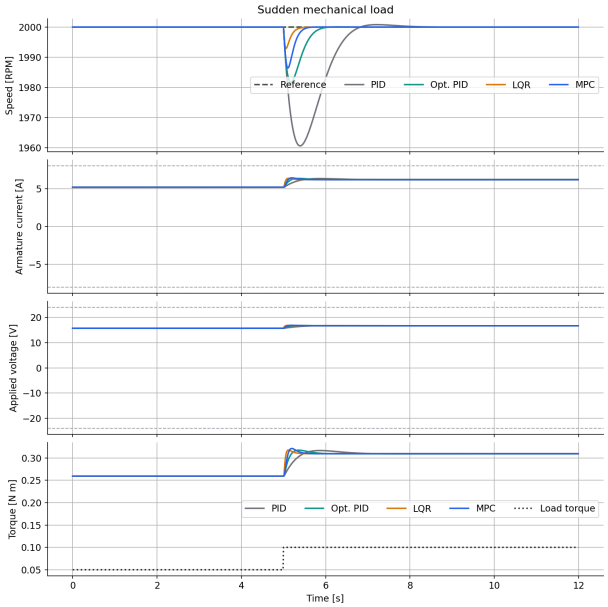


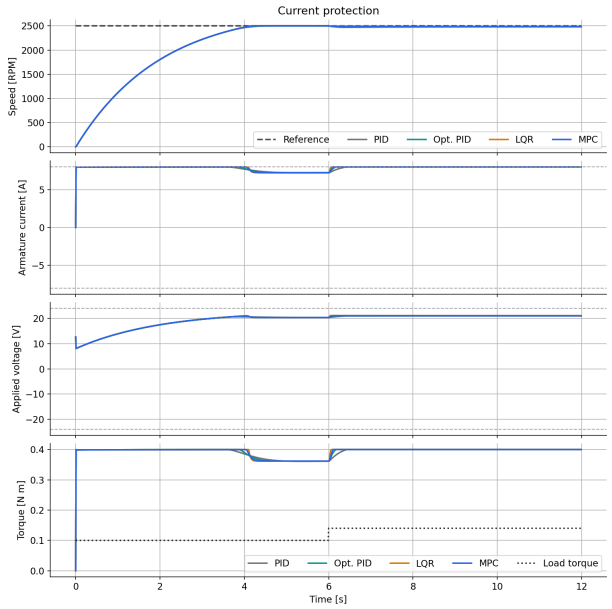
Figure 2: IAE by scenario.



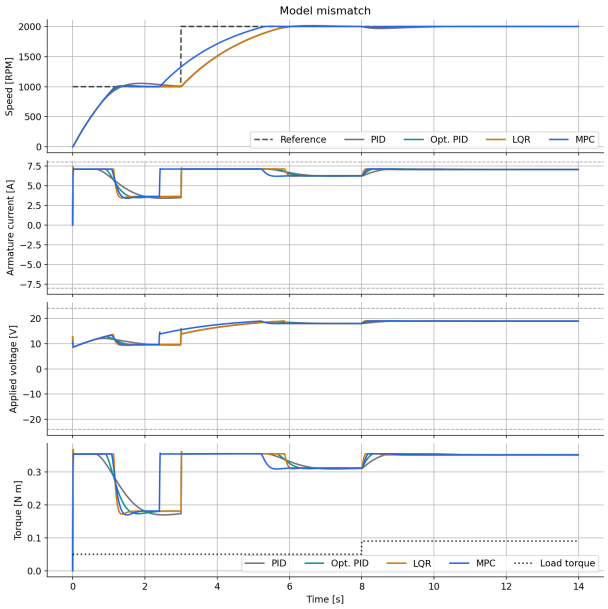
(a) Speed tracking



(b) Sudden load



(c) Current protection



(d) Model mismatch

Figure 3: Speed, armature current, voltage and torque for all four controllers.

6.1 Tracking and load changes

LQR settles startup in 0.71 s and MPC in 0.76 s, against 1.84 s for the hand-tuned PID and 0.92 s for the optimised PID. MPC has the lowest IAE on speed tracking (1071 against 1854, 1660 and 1608 for PID, optimised PID and LQR), helped by reference preview. On the load step the LQR is lowest (IAE 1.2 against 3.2 for MPC, 8 and 35 for the PIDs), and on the variable load too (18 against 45, 96 and 275): current feedback plus an integrator reacts as fast as the MPC's disturbance estimator. A PID with a larger integral gain could narrow the gap to both, at the cost of noise sensitivity.

6.2 Drive limits

In the voltage-saturation and current-protection scenarios the actuator, not the algorithm, sets the best achievable response, and the controllers end up close (current-protection IAE 3626 for MPC against 3678, 3635 and 3616). MPC keeps the drive out of its current limit for less time (4.34 s

against 9.18, 9.71 and 10.05 for the LQR), because it plans along the 8 A bound instead of running into it; the LQR has no knowledge of the limit. Under ±20% resistance error the drive's current limit can be exceeded transiently, up to 9.1 A against an 8 A limit in the low-resistance corner, before the feedback term backs off.

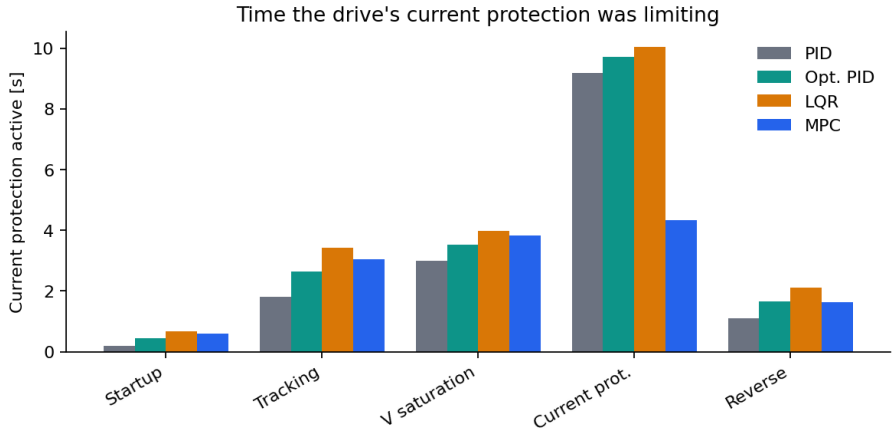


Figure 4: Time the drive's current protection was limiting, in the scenarios that reach the limit.

6.3 Noise and robustness

With 15 RPM encoder noise the MPC keeps the lowest IAE (103 against 181, 160 and 173 for the LQR), after the observer gain was lowered to 0.1; at gain 1 it amplified the noise into the voltage and lost to the PID. Under plant mismatch the MPC stays offset-free and has IAE 1290 against 1700 for the frozen optimised PID, 1706 for a PID retuned on the mismatched plant and 1676 for the LQR (Figure 5). Execution time differs by orders of magnitude: about 0.01 ms per sample for the PID loops and LQR, 3–6 ms for MPC.

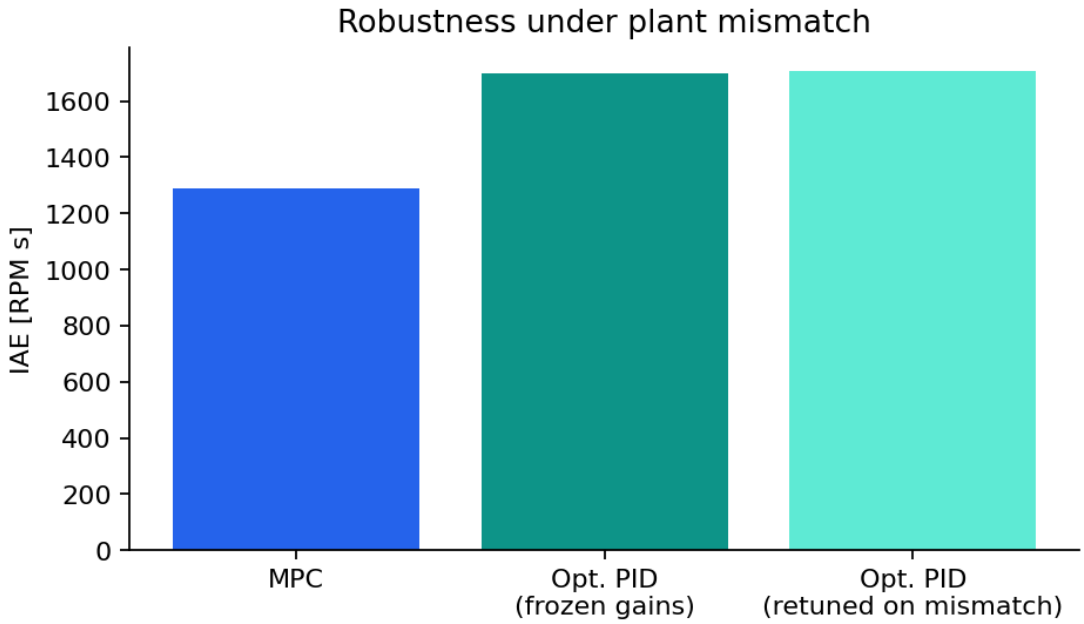


Figure 5: IAE under plant mismatch: MPC, frozen optimised PID, and optimised PID retuned on the mismatched plant.

7 Discussion and limitations

Offset-free MPC is the best overall here, but the comparison favours it and the LQR in three ways: both use the measured current and a nominal model that matches the plant when no mismatch is applied, the PID loops are single-loop speed controllers, and the MPC weights, horizon and observer gain and the LQR weights are hand-set after looking at these scenarios. The optimised PID is tuned on one scenario, so its advantage elsewhere is whatever carries over, which is the point of freezing it; retuning on the mismatched plant did not help it.

The model is linear with constant parameters. It omits brush drop, magnetic saturation, temperature dependence of the resistance, Coulomb friction and shaft compliance; the load torque keeps its sign at standstill; the current sensor is ideal; and the converter efficiency (0.95) is illustrative, so the drive and motor efficiency figures in the repository (about 0.1–0.2, dominated by copper loss at these small shaft powers) are not data. No stability proof is claimed for the MPC or for the LQR under saturation, and the solve time of a few milliseconds is not a hard real-time guarantee.

8 Conclusion

For this PMDC motor drive, offset-free MPC with reference preview is the controller to prefer when constraints, previewed references and disturbances matter: it leads on tracking, noise, reversal and mismatch and plans along the current limit instead of hitting it, at the price of a numerical optimisation every sample. An integral-augmented LQR is a cheap and strong alternative for load disturbances and startup, but it does not respect the current limit by itself. Where the drive's limits decide the outcome, the controllers are close, and a well-tuned PID remains a reasonable choice.

References

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